

# Efficient, Ultra-flat and Low-noise Parametric Comb Generation using Hybrid Nonlinear Fibre System and Chirped-pulse Nonlinear Loop Mirror

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**Abstract** We address efficiency, spectral flatness and noise-scaling problems in cavity-less frequency comb generation using a hybrid fibre system combining soliton and wave-breaking nonlinearities. We propose chirped-pulse nonlinear loop mirror to efficiently shape pulse and spectra, achieving 3-dB flatness across C-band and sub-3-kHz linewidth over 80nm. ©2026 The Author(s)

## Introduction

Optical frequency combs with 10s of GHz tone spacing have been exploited to replace laser banks in wavelength-division multiplexing (WDM) optical communications, offering potential advantages including more efficient nonlinear mitigation [1,2], efficient phase noise compensation [3], and reduced cost [4] due to the ease of thermal and wavelength control. Beyond transmission, low-noise and flat-spectrum comb sources have demonstrated advantages in clock and RF synchronization [5,6], photonic analogue-to-digital conversion [7], and signal processing [8].

Modern communication systems require low-noise, flat-spectrum sources to support high-capacity transmission. As WDM communications expands beyond the telecom C-band, broad bandwidth and tuneable channel spacing become important to ensure sufficient power per tone for modulation. Further, energy efficiency and robustness against environmental perturbations are critical for practical field deployment.

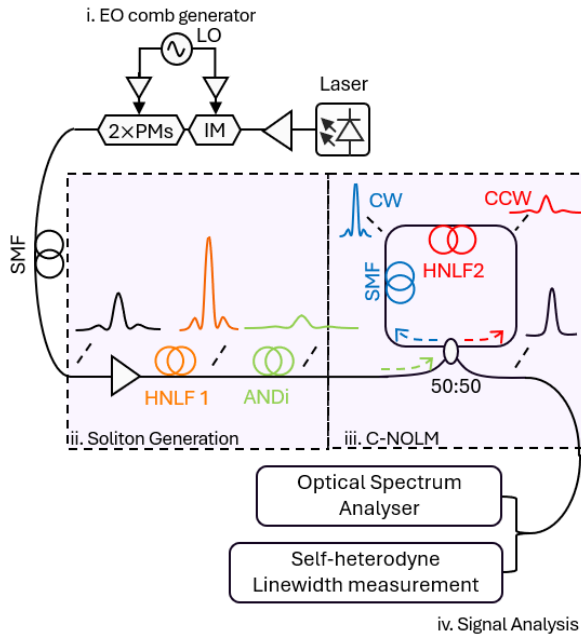
Optoelectronic combs offer robust operation and tuneable tone spacing [9,10]. When combined with parametric spectral expansion, they have demonstrated 10 dB bandwidth exceeding 100 nm [11,12]. However, achieving ultra-flat spectral profiles requires existing approaches to employ multiple stages of nonlinear optical/amplifying loop mirrors (NOLM/NALM) [13,14] or free-space-based pulse/spectrum shapers [15] to improve the flatness from ‘bat-ear’ aberrations. These solutions are bulky, costly, inflexible and difficult to integrate. In addition, existing approaches predominantly rely on normal-dispersion highly nonlinear fibre to suppress noise arising from modulation instability. However, this inevitably requires long fibre lengths, which introduce additional noise due to fibre birefringence and the noise accumulation as signals propagate along the fibre.

In this paper, we propose and demonstrate a hybrid fibre system for parametric comb expansion that enables significantly improved comb generation efficiency and spectral flatness, while maintaining high optical signal-to-noise ratio (OSNR) and low linewidth. These performance improvements are achieved through two key innovations in the fibre system design. The first innovation is the optimal combination of anomalous- and normal-dispersion highly nonlinear fibres, which enables high-peak-power soliton generation over a short fibre length in a highly coherent process. The resulting soliton-like pulse is then launched into the chirped-pulse nonlinear loop mirror (C-NOLM) [16] fibre system, which is the second innovation, where pedestal suppression efficiently reshapes the pulse into a quasi-Gaussian temporal and spectral profile with a 3 dB bandwidth spanning the entire C-band.

Compared to prior art which requires cascading NOLM and NALM for pulse and spectrum shaping, we achieve a spectral flatness of 3 dB over telecom C-band using a single C-NOLM. Instead of employing 50-70 metres of highly nonlinear fibre and bi-directional amplifiers in a NALM design, which introduce power fluctuation, our method significantly reduces the total fibre length to less than 10 meters, thereby significantly reducing fibre-induced noise. This results in an  $\text{OSNR}_{0.1\text{nm}}$  exceeding 33 dB, sub-3kHz linewidth over 1520-1585nm, and 220-fs pulse with a Gaussian shape.

## Experiment Setup

Fig.1 shows the experimental setup. Our cavity-less parametric comb includes three stages: (i) a single pass electro-optic (EO) frequency comb, (ii) a hybrid-fibre soliton compression stage and (iii) a C-NOLM based pulse/spectrum shaper. The first stage is a standard EO comb starting with a 1.6-kHz-linewidth continuous-wave laser operation at 1555 nm. It is amplified



**Fig. 2:** Experimental setup for the comb generator. The conceptual pulse-evolution diagram is shown alongside the system schematic.

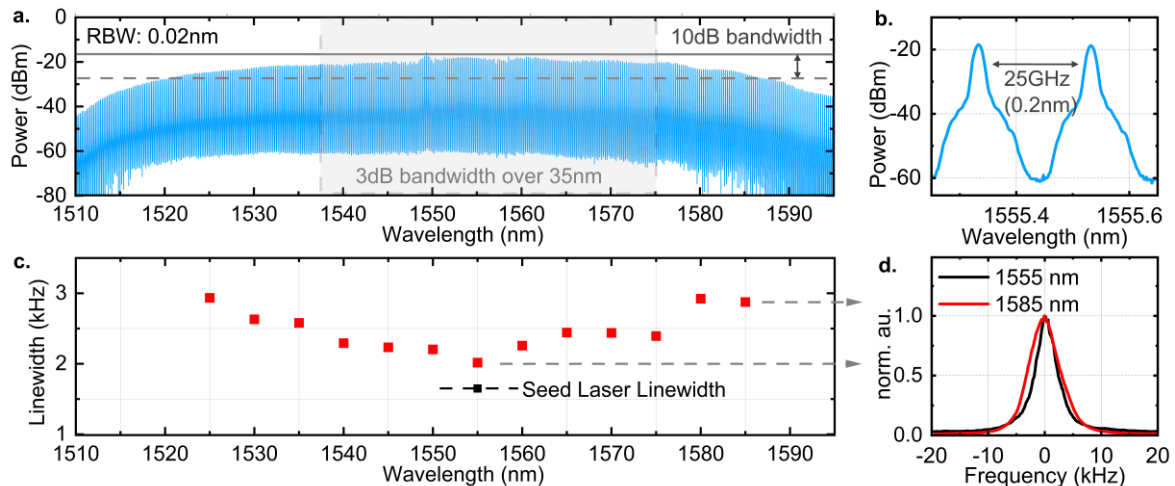
to 0.5 Watt before seeding a cascade of intensity and phase modulators driven at 25-GHz RF signals, outputting a comb signal with a bandwidth of approximately 10 nm. The comb signal is a quasi-linearly chirped pulse in the time domain and is subsequently compressed to about 700 fs full-width at half-maximum (FWHM) using 70 metres of PM single-mode fibre. This generates a pulse train with an average power of 6 dBm and the peak-to-pedestal ratio of about 10 dB [12].

The second stage, as shown in Fig.2 (ii), is a soliton compression stage, where the 700-fs pulses are amplified to 0.5 Watt and launched into a hybrid fibre system. The first segment of the fibre system is a 13-m highly nonlinear fibre

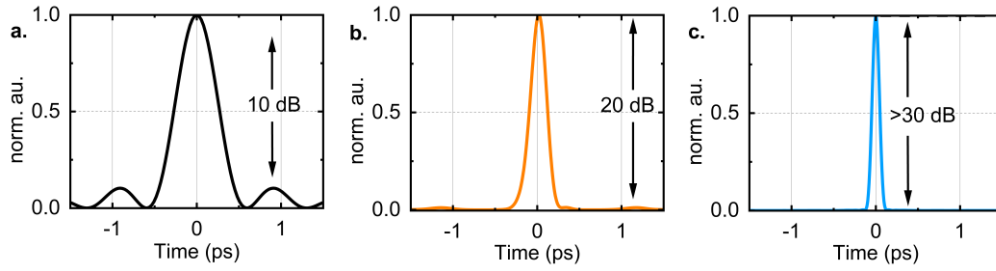
(HNLF1) with an anomalous dispersion of  $+5 \text{ ps/nm/km}$  and a nonlinear coefficient (effective gamma) of  $11 \text{ W}^{-1}\text{km}^{-1}$ . As the pulses propagate through this segment, they are compressed to a 150-fs pulse with a peak power of about 75 W at the end of the segment. Following, the soliton-like pulse is launched into 3-m all normal dispersion (ANDi) fibre, which has a dispersion of  $-50 \text{ ps/nm/km}$  at 1550 nm and is spliced to the HNLF1 with a 1 dB loss, to stretch the pulse.

The C-NOLM is introduced in the last stage, as shown in Fig. 2 (iii). Different from conventional NOLM, which requires unequal splitting ratios (e.g. 40:60 in [13]) or an in-loop attenuator to create power asymmetry into the loop, our C-NOLM uses 50:50 splitting and relies on asymmetry created by the ordering of dispersive and nonlinear elements in the loop for pulses propagating in clockwise (CW) and counterclockwise (CCW) directions.

The clockwise propagating pulse first passes through a 5 m PM fibre that compensates for the dispersion introduced by the ANDi fibre. As a result, a short and high peak power pulse (shown in blue) is formed before entering HNLF2, which is a 9 m HNLF with a dispersion of  $-0.5 \text{ ps/nm/km}$  and an effective gamma of  $11 \text{ W}^{-1}\text{km}^{-1}$ . Due to the high peak power, the Kerr nonlinearity induces an intensity dependent phase shift onto clockwise propagating pulses. In contrast, the CCW pulses propagate through HNLF2 in their stretched state, as shown in Fig. 2(iii) in red, with a low peak power of about 5 W. As a result, they do not undergo a self-phase-modulation-induced phase shift and are subsequently compressed by the SMF. Consequently, the CW and CCW pulses retain the same pulse shape after passing through the loop and interfere at the coupler. The SPM-



**Fig. 1:** (a) C-NOLM output spectrum. (b) Detailed spectrum around 1555.5 nm. (c) Seed laser linewidth (black square) and measured linewidths across the spectrum (red squares). (d) Example self-heterodyne beating signal measured at 1555nm (black trace) and the corresponding fitting profile (red trace).



**Fig. 3:** Simulation pulse shape comparing the performance of a conventional NOLM and our proposed C-NOLM. (a) Compressed pulse from EO comb. (b) NOLM output assuming a classic 60:40 split. (c) Output of our proposed C-NOLM.

induced phase difference, which occurs mainly at the centre of the pulses where the power is highest, enables the transmission of Gaussian-like pulses, while the pulse wings (pedestal) are reflected.

The spectra of comb signals are analysed using an optical spectrum analyser (OSA) with a resolution of 0.02 nm. We also characterise line-by-line linewidth using self-heterodyne method after filtering out each comb line. The pulse shapes are estimated by assuming Fourier transform limit.

## Results and discussions

Fig. 2(a) shows the optical spectrum of our obtained comb at the output of the C-NOLM. We highlight less than 3-dB spectrum flatness across the telecom C-band (1535-1565 nm), which has not been achieved previously with any single-stage time-domain pulse shaping strategies, making our comb a promising source for coherent transmissions. We measure a 10-dB bandwidth of 75 nm (1520-1595 nm), which is limited by the pulse width. This can potentially be further expanded to more than 100 nm with another expansion stage, as demonstrated in [12]. Fig. 2(b) illustrates a zoom-in spectrum at 1555.5nm showing a tone spacing of 25 GHz and an  $\text{OSNR}_{0.1\text{nm}}$  exceeding 33 dB. Fig. 2(c) plots the measured line-by-line linewidths. Benefitting from a highly coherent design with short nonlinear fibre length, the linewidth only slightly increased from 2 kHz to about 3 kHz as the wavelength deviates from the centre (1555 nm) to the edge (1525 nm and 1585 nm). Fig. 2(d) shows an example self-heterodyne measurement comparing centre tone at 1555 nm (black) and an edge tone at 1585 nm (red), acquired with a 100  $\mu\text{s}$  measurement time.

As pulse shape can be used to estimate the spectrum shape and flatness, we compare the time domain pulse shapes of the compressed EO comb signal (Fig.3a), conventional NOLM based on 60:40 coupler (Fig.3b), and our proposed hybrid fibre enhanced C-NOLM (Fig.3c). As discussed in previous literature [12], the pedestal of the pulses contributes to the spectrum variation

| Comparison               | NOLM  | Our C-NOLM |
|--------------------------|-------|------------|
| Pulse width              | 240fs | 130fs      |
| Peak to pedestal Ratio   | 20dB  | 30dB       |
| C-band Spectrum flatness | 25dB  | 3dB        |

**Table 1:** NOLM and C-NOLM comparison.

particularly near the centre of the spectrum.

Generally, a higher pedestal suppression ratio leads to a flatter spectrum. Fig.3 (a) shows the simulated pulse shape of the 1<sup>st</sup> stage EO comb after compression to the transform limit, reconstructed from the measured spectrum and exhibiting a peak-to-pedestal ratio of approximately 10 dB. A conventional NOLM, which typically requires a 60:40 coupler with HNLF of 70 m, as shown in [11], can enhance the peak-to-pedestal ratio to approximately 20 dB, as our simulation shows in Fig. 3 (b). Our proposed hybrid fibre-based C-NOLM, however, can significantly enhance peak to pedestal ratio to 30 dB in a single stage design, as shown in Fig. 3 (c).

Table 1 compares the key performance indicator. Using the same seed source, our approach significantly reduces the pulse width to 130 fs, as opposed to 240 fs using a NOLM. We also achieved 10 dB higher peak to pedestal ratio as well as 3 dB spectral flatness within telecom C-band, whilst NOLM could only slightly broaden the spectrum with 25 dB power variation.

## Conclusions

We propose and demonstrate a cavity-less frequency comb generator based on a hybrid-fibre C-NOLM. The fibre system significantly improves the efficiency and performance compared to conventional nonlinear parametric comb expansion and spectral/pulse shaping approaches, achieving <3 dB spectral flatness over telecom C-band, >33 dB OSNR and sub-3 kHz linewidth over an 80 nm spectral width using a single shaping stage with <10 metres of nonlinear fibres. Our results provide a robust, integrable platform for developing high-performance comb sources.

## Acknowledgements

The authors acknowledge EPSRC grant ORBITS (EP/V051377/1), TRANSNET (EP/R035342/1) and EPSRC fellowship PROSPECT (EP/Z534444/1).

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